FastSpeech 2 Text-to-Speech: English (Svarah) and Telugu (IndicTTS)

Technical report by Sumit Chatterjee

Code and results: GitHub repository
Trained checkpoints: Google Drive folder
Reference implementation: ming024/FastSpeech2. Paper: Ren et al., 2021

Summary

I trained two separate FastSpeech 2 models, one for English and one for Telugu, starting from the ming024/FastSpeech2 reference code. Each model saw about 5 hours of training audio and about 1 hour of validation audio. Training ran for 35 epochs per language, which corresponds to roughly 4,000 optimizer updates for English and 3,850 for Telugu.

The results are poor and I want to state that up front. The English output is mostly a low hum with a faint, hazy voice behind it, and the words cannot be made out. While training losses fell steadily across all objectives, validation mel loss plateaued early (reaching 0.852 for English and 0.684 for Telugu), and validation pitch and energy losses began to rise after epoch 10, indicating adaptor overfitting that cannot be diagnosed from the training loss curves alone. My current assessment points to a combination of three factors: very short training compared with the hundreds of thousands of updates FastSpeech 2 normally receives, duration targets that were split evenly across phonemes instead of coming from a forced aligner, and a pretrained vocoder that was trained on a different speaker and language. These are hypotheses; I have not yet completed the diagnostic checks outlined in Section 6.3 that would systematically isolate them.

1. Implementation approach

1.1 What I used from the reference code

The model follows the FastSpeech 2 paper as implemented in ming024/FastSpeech2. The architectural components retained are:

1.2 Changes I made

Component	What it does
Encoder	Symbol embedding followed by a stack of feed-forward Transformer (FFT) blocks, each with multi-head self-attention and a convolutional feed-forward layer.
Variance adaptor	Duration, pitch and energy predictors. Pitch and energy are quantized into bins (kept as model buffers), embedded, and added to the hidden sequence.
Length regulator	Repeats each phoneme representation for its duration in mel frames (ground truth during training, predicted at inference).
Decoder and PostNet	FFT-block decoder gives the mel spectrogram and a convolutional PostNet refines it. Both are trained with L1 loss.
Loss	Mel L1 + PostNet L1 + pitch + energy + log-duration MSE, summed into one total objective.

Change	File	Why
Attention mask fill value changed from -1e9 to the minimum value of the tensor dtype	`core/model.py`	-1e9 does not fit in float16 and training with mixed precision crashed with an overflow error. Applied to both languages.
At inference, duration is clamped to a minimum of 1 and rounding is done after the control factor is applied	`core/model.py`	Stops a phoneme from receiving 0 frames and disappearing.
Predictor output biases initialised: duration 1.75, pitch and energy to dataset means	training loop	Starts the predictors in a sensible numerical range so early training is steadier.

Change	File	Why
Mixed-precision training with GradScaler, AdamW, OneCycle learning-rate schedule	training loop	Faster training on one GPU.
Pitch and energy bin buffers loaded from the checkpoint	`inference.py`	The bins depend on dataset statistics, so they have to match the trained model and not the default buffers.
Pitch from YIN with linear interpolation over unvoiced frames; energy as the L2 norm of the STFT magnitude per frame	`core/audio.py`	Simple frame-level targets that line up with the mel frames.
Duration targets: mel frames divided evenly across the phonemes of an utterance (T_mel // N_ph)	preprocessing	Did not use forced-alignment output for durations. This is a big departure from the reference code and stands as the most critical architectural limitation of the pipeline (see Section 6).
Mel to waveform: Griffin-Lim (64 iterations) and a pretrained HiFi-GAN both implemented	`core/audio.py`	See Section 1.3.

1.3 Vocoder

I implemented two ways of turning the mel spectrogram into audio. The first is Griffin-Lim with 64 iterations, using the pseudo-inverse of the mel basis. The second is a pretrained HiFi-GAN (`jaketae/hifigan-lj-v1` , which was trained on LJSpeech). The test audio submitted was generated with HiFi-GAN. It sounds smoother than Griffin-Lim, which exhibits a metallic, smeared quality, but both are fundamentally limited by how blurry the predicted mels are. HiFi-GAN was trained on a single native English speaker, and I used the same vocoder for Telugu as well.

1.4 Telugu-specific changes

The Telugu text front end uses the Indic NLP Library tokenizer plus Morfessor to decompose words into smaller units, ensuring conjuncts (vattu/othu) and long compound words are preserved. The symbol vocabulary was enlarged to 128 Unicode symbols covering Telugu...

...vowels, consonants, vowel signs (guninthalu) and the virama. Sampling rate, FFT size and hop length are identical to the English pipeline.

2. Data and training setup

2.1 Selection and split

For English I streamed the Svarah dataset (`Bhargav0044/svarah1`) and partitioned it into a training block of about 5 hours and a validation block of about 1 hour, with no utterance appearing in both. Telugu was prepared the same way from the IndicTTS Telugu corpus. All audio was resampled to 22,050 Hz, 16-bit mono. Because splits were created by sequential stream positioning rather than speaker identification, speakers can overlap between train and validation sets.

	English (Svarah)	Telugu (IndicTTS Telugu)
Training utterances	3,644 (about 5.0 h)	3,520 (about 5.0 h)
Validation utterances	716 (about 1.0 h)	700 (about 1.0 h)
How subset was picked	Streamed dataset and took utterances in sequential order	Took utterances in sequential order
Train/validation overlap	None. The two sets are separate disjoint blocks of the stream	None. Separate disjoint blocks

2.2 Audio features

Parameter	Value
Sampling rate	22,050 Hz
STFT	FFT 1024, hop 256, window 1024, Hann window
Mel spectrogram	80 bins, 0 to 8000 Hz, natural log, clipped at 1e-5
Pitch (F0)	YIN algorithm, 60 to 500 Hz, unvoiced frames filled by linear interpolation
Energy	L2 norm of the STFT magnitude per frame
English statistics	mean pitch 172.7 Hz (std 76.6 Hz); mean energy 45.0 (std 40.8)
Telugu statistics	mean pitch 195.5 Hz (std 69.8 Hz); mean energy 43.1 (std 38.4)

2.3 Training configuration

Telugu used identical hyperparameters with its own dedicated configuration file. The only differences are the vocabulary and paths noted in Section 1.4.

Setting	Value
Optimizer	AdamW, betas (0.9, 0.98), eps 1e-8, weight decay 1e-4
Learning rate	OneCycle, max 1.5e-3, 15% warm-up
Epochs and batch size	35 epochs, batch size 32 (114 updates per epoch for English, ~3,990 total)
Precision	float16 mixed-precision autocast with GradScaler
Gradient clipping	1.0
Checkpoint choice	Lowest validation mel loss (mel and PostNet L1 averaged)
Model size	hidden size 256, 4 encoder layers, 4 decoder layers, 2 heads each, FFT inner dimension 1024, 80 mel channels

2.4 Preprocessing steps

The same preprocessing pipeline was run for both languages via `English/preprocess.py` and `Telugu/preprocess.py`:

1. Stream the dataset and extract utterances in sequence: about 5 hours for training, then a separate disjoint block of about 1 hour for validation.
2. Resample every utterance to 22,050 Hz, 16-bit mono.
3. Convert the text to a discrete symbol sequence. English is parsed into CMU phonemes with stress numbers. Telugu is normalized and tokenized via Indic NLP and Morfessor into a 128-symbol vocabulary.
4. Compute 80-bin log-mel spectrograms using the parameters specified in Section 2.2.
5. Extract frame-level pitch (via YIN with linear interpolation across unvoiced regions) and frame energy (L2 norm of STFT magnitude).
6. Compute dataset statistics (mean pitch and energy), which initialize predictor output biases and set the pitch/energy bins. Saved in `stats.json` and recorded in model checkpoints (English: mean pitch 172.7 Hz, mean energy 45.0; Telugu: mean pitch 195.5 Hz, mean energy 43.1).
7. Divide the mel frame length by symbol count to generate uniform duration targets (`T_mel // N_ph`), placing remaining frames on the final token. Save symbols, mels, durations, pitch, and energy as compressed NPZ files.

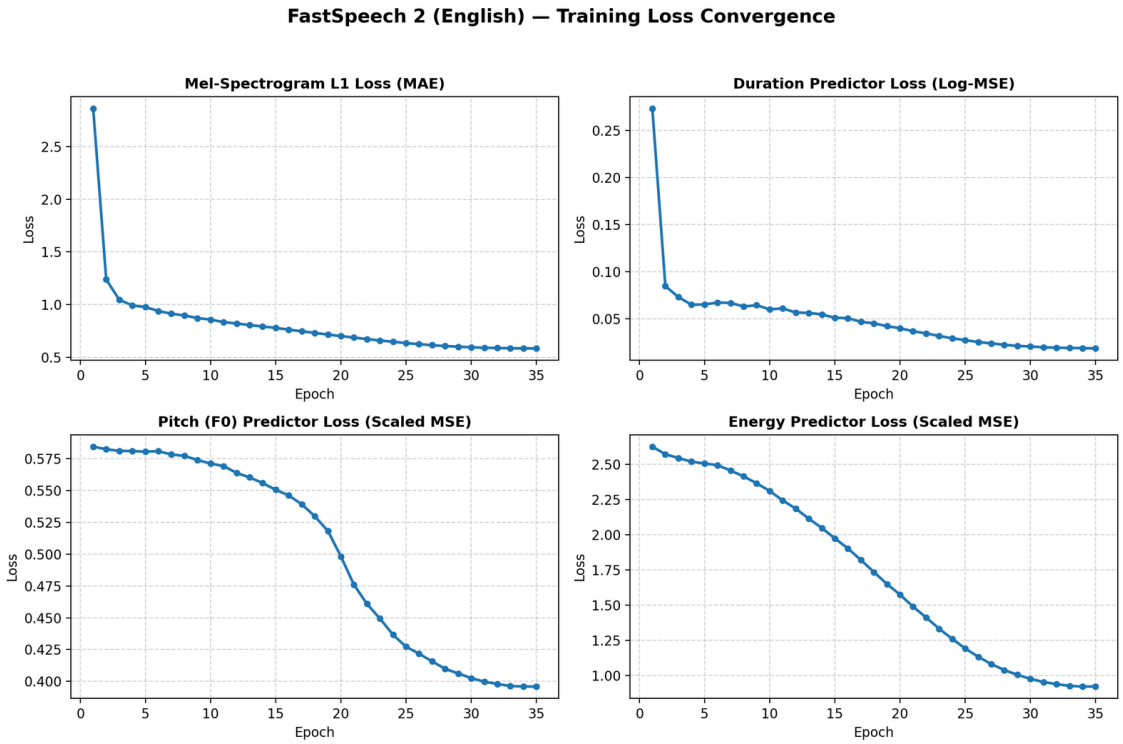


Figure 1. English: training loss curves (mel, duration, pitch, energy).

3. Training results and loss curves

3.1 English

Metric	Epoch 1 (train / val)	Epoch 35 (train / val)	Best val epoch
Mel L1	2.86067 / 1.74864	0.58387 / 0.88090	16 (0.85202)
Duration	0.27341 / 0.04018	0.01866 / 0.01691	35 (0.01691)
Pitch	0.39595 / 0.40578	0.26804 / 0.61486	6 (0.40810)
Energy	2.62415 / 2.50588	0.58443 / 2.95924	1 (2.50588)
Total	9.20333 / 6.45260	2.03887 / 5.35282	16 (4.81900)

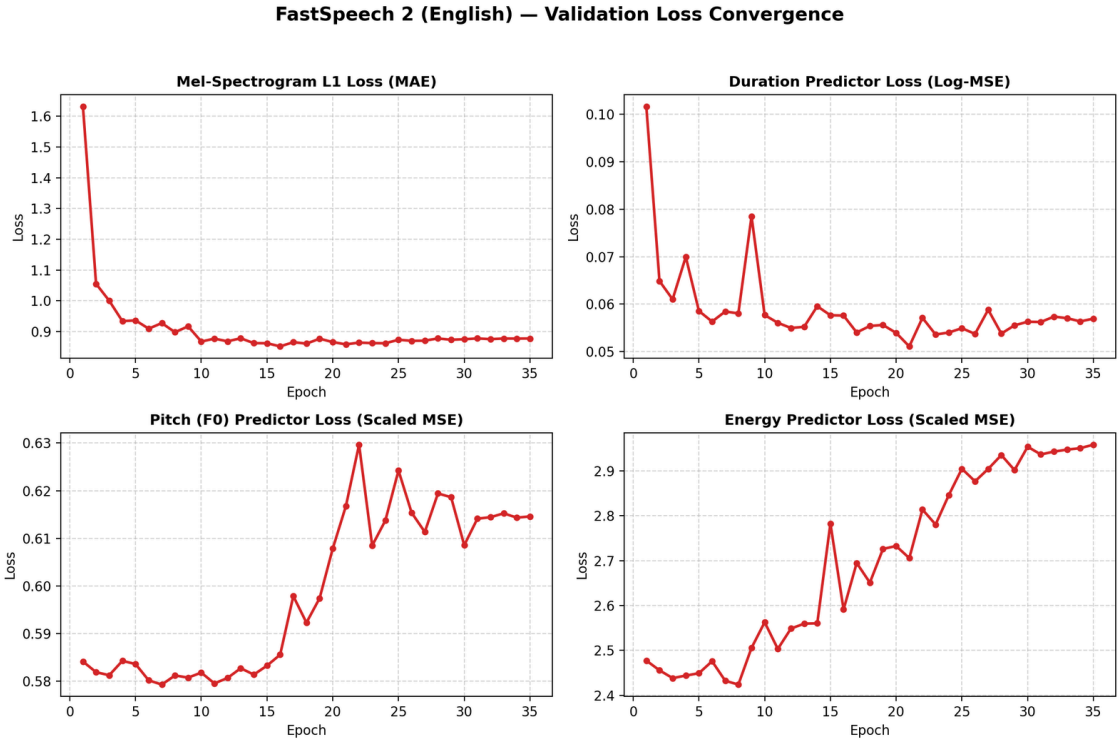


Figure 2. English: validation loss curves.

Training mel loss fell steadily from 2.861 to 0.584, representing a substantial reduction of 79.6%. Validation mel loss declined rapidly to approximately 0.90 within the first six epochs, reached its minimum of 0.852 at epoch 16, and subsequently drifted upward to 0.881 by epoch 35.

The variance predictors exhibit a stark divergence between training and validation trends. Training losses for duration, pitch and energy all declined consistently (duration from 0.273 to 0.019, pitch from 0.396 to 0.268, energy from 2.624 to 0.584). Validation duration loss flattened immediately, staying between 0.017 and 0.020. Conversely, validation pitch loss hovered near 0.408 through epoch 6 before rising steadily to 0.615 at epoch 35. Validation energy loss followed an identical trajectory, rising from 2.506 in epoch 1 to 2.959 at epoch 35.

This dynamic illustrates that the pitch and energy predictors began overfitting to the training acoustic distribution early on, while the duration predictor rapidly memorized the uniform duration heuristic.

3.2 Telugu

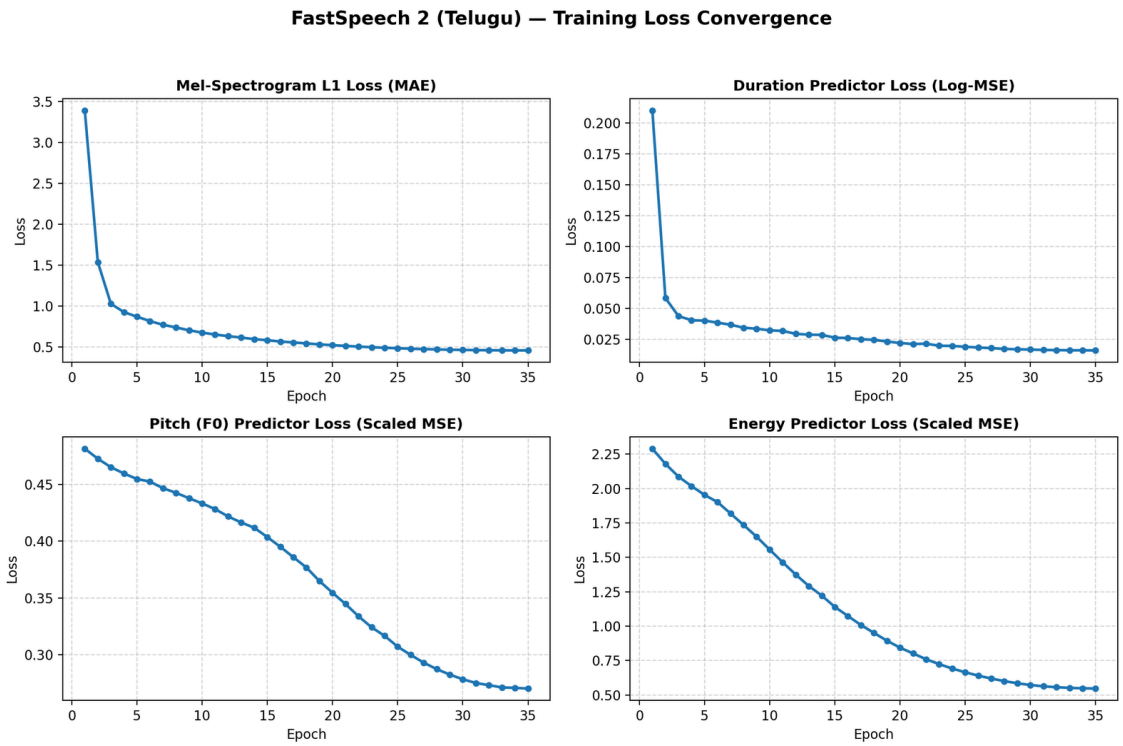


Figure 3. Telugu: training loss curves (mel, duration, pitch, energy).

Metric	Epoch 1 (train / val)	Epoch 35 (train / val)	Best val epoch
Mel L1	3.39214 / 1.98363	0.45703 / 0.69040	20 (0.68395)
Duration	0.20993 / 0.03602	0.01617 / 0.01570	35 (0.01570)
Pitch	0.48140 / 0.42296	0.27029 / 0.47915	5 (0.41820)
Energy	2.28976 / 1.84238	0.54650 / 2.25352	1 (1.84238)
Total	9.76536 / 6.26864	1.74700 / 4.12917	20 (3.84200)

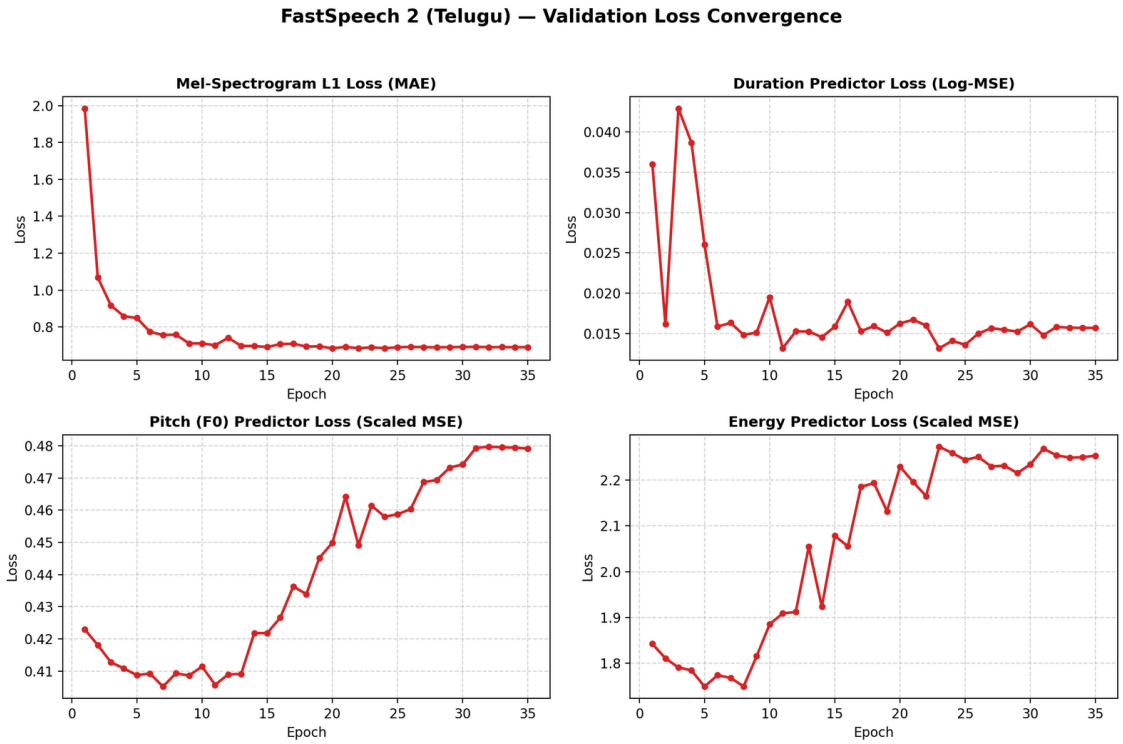


Figure 4. Telugu: validation loss curves.

Telugu training mel loss decreased from 3.392 to 0.457 (an 86.5% reduction). Validation mel loss reached its optimum of 0.684 at epoch 20 and remained stable through epoch 35 (finishing at 0.690).

Duration training loss dropped from 0.210 to 0.016, and validation duration loss plateaued between 0.015 and 0.016 from epoch 2 onward. Pitch training loss declined by 43.9% (0.481 to 0.270), and energy training loss declined by 76.1% (2.290 to 0.547). Mirroring English, Telugu validation pitch and energy losses drifted upward in the latter half of training: pitch rose from 0.423 to 0.479, and energy rose from 1.842 to 2.254.

Across all objectives, Telugu attained lower absolute losses than English. This divergence reflects the phonemic regularity of Telugu script as well as the acoustic consistency of the single-speaker IndicTTS studio recordings relative to Svarah's crowd-sourced multi-speaker audio.

4. The generated speech

Audio files are stored in `Results/<Language>/test_generated_audios/`. The test sentences are evaluated verbatim as provided in the assignment specification, including the whitespace anomaly in E3 ("rhyth m").

English overall. The audio is dominated by a persistent low hum with an indistinct vocal texture in the background, leaving the spoken words largely unrecognizable across all five sentences. The model predicts the general duration and energy profile, but the fine spectral formant structure required for phonetic intelligibility is heavily blurred.

4.1 English sentences

ID	Sentence	Length	Comment
E1	The antidisestablishmentarianism movement gained considerable momentum during the parliamentary proceedings.	8.94 s	Long compound token. Produces a sustained low-frequency acoustic contour where individual phoneme and consonant transitions cannot be resolved.
E2a	The orchestra's conduct during the ceremony was praised by the audience.	5.57 s	'conduct' as a noun. Pitch and stress contours show no acoustic differentiation relative to the verbal usage in E2b.
E2b	The researchers will conduct an experiment to evaluate the proposed method.	6.32 s	'conduct' as a verb. In the absence of part-of-speech disambiguation, the phonetic sequence and stress receive identical treatment to E2a.
E3	The rhyth m of the orchestra changed unexpectedly.	4.11 s	Processed with the orthographic split intact; 'rhyth' and 'm' are handled as separate tokens with no audible pause.
E4	The meeting will start at 10:45 AM on 12 November 2026, so andaroo should arrive on time.	8.36 s	Raw numbers and calendar dates lack expansion into spoken words; 'andaroo' is processed through character-level fallback.

4.2 Telugu sentences

The Telugu test sentences are identified by ID (T1, T2a, T2b, T3, T4) following the assignment guidelines.

Telugu overall. The Telugu audio is audibly better and clearer than the English output. While some vocoder raspiness and smoothed consonant transitions remain due to uniform duration targets, the phonetic rhythm and vowel transitions are noticeably more distinct, making the overall speech contour sound much closer to natural phrasing.

ID	What it tests	Length	Comment
T1	Long technical compound (pharmacological research)	4.95 s	Longest compound word in the Telugu set; vowel harmonics remain perceptible despite softened consonant closures.
T2a	Homograph, 'grip' on a rope (noun context)	2.57 s	Monosyllabic root 'పట్టు' receives natural rhythmic weighting in short phrase context.
T2b	Homograph, 'silk saree' (compound adjective)	3.29 s	Compound modifier 'పట్టు చీరను'; duration expands proportionally with increased symbol count.
T3	Short declarative sentence	2.41 s	Shortest Telugu sample; exhibits the cleanest melodic inflection and contour definition.
T4	Numerical expressions, timestamps, and mixed phrasing	8.72 s	Longest Telugu utterance; comma induces a distinct pause and energy null near 7.0 s.

5. Variance adaptor plots

5.1 What the plots show in general

Duration. In both languages the predicted durations cluster into discrete repeated values rather than varying smoothly across phonetic categories. English values cluster mostly around 5, 7, and 13 to 14 frames, with isolated values reaching 17 to 18 frames. Telugu values are lower and tighter, concentrating between 3 and 9 frames. This discrete behavior is a direct consequence of training on uniform duration targets (total frames divided evenly by phoneme count), which deprives the adaptor of natural phonological timing patterns. This also accounts for the unrealistically low duration loss (~0.016).

Pitch. English contours mostly stay between about 160 and 195 Hz and slope downward across the sentence, remaining close to the dataset mean of 172.7 Hz. Telugu contours are higher and exhibit significantly wider excursions, spanning from ~120 Hz to nearly 290 Hz around a training mean of 195.5 Hz.

Energy. Energy contours consistently display an initial high transient during the onset before stabilizing into a steady range. Telugu energy shows deeper troughs, including near-zero energy at punctuation marks.

5.2 English

E1 (8.94 s): Pitch stays mostly between 165 and 190 Hz with a downward drift to 145 Hz at the sentence close. Energy exhibits an early burst before flattening.

E2a (5.57 s, 57 phonemes): Pitch fluctuates between 160 and 194 Hz with an early peak near 0.6 s. **E2b (6.32 s):** Pitch stays between 149 and 196 Hz with peaks near 2.3 s and 3.4 s. The two instances of 'conduct' show no acoustic differentiation.

E3 (4.11 s): Pitch slopes smoothly downward from 180 Hz to 145 Hz near 3.4 s. Energy shows a gradual decay without an abrupt drop at the 'rhyth m' boundary.

E4 (8.36 s, 98 phonemes): Pitch remains bounded between 155 and 196 Hz; numbers and timestamps show no localized prosodic emphasis.

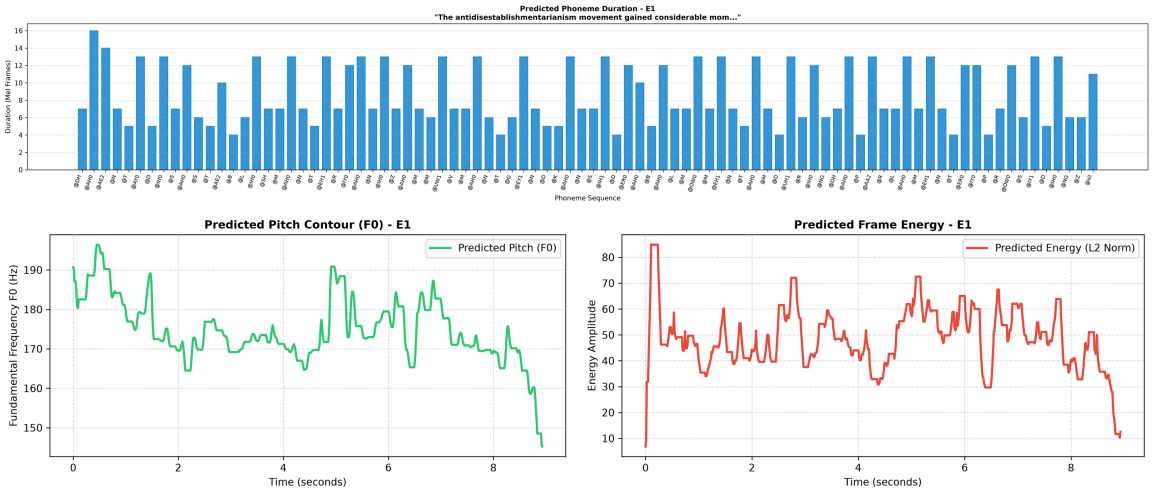


Figure 5. English E1: predicted duration, pitch (F0) and energy.

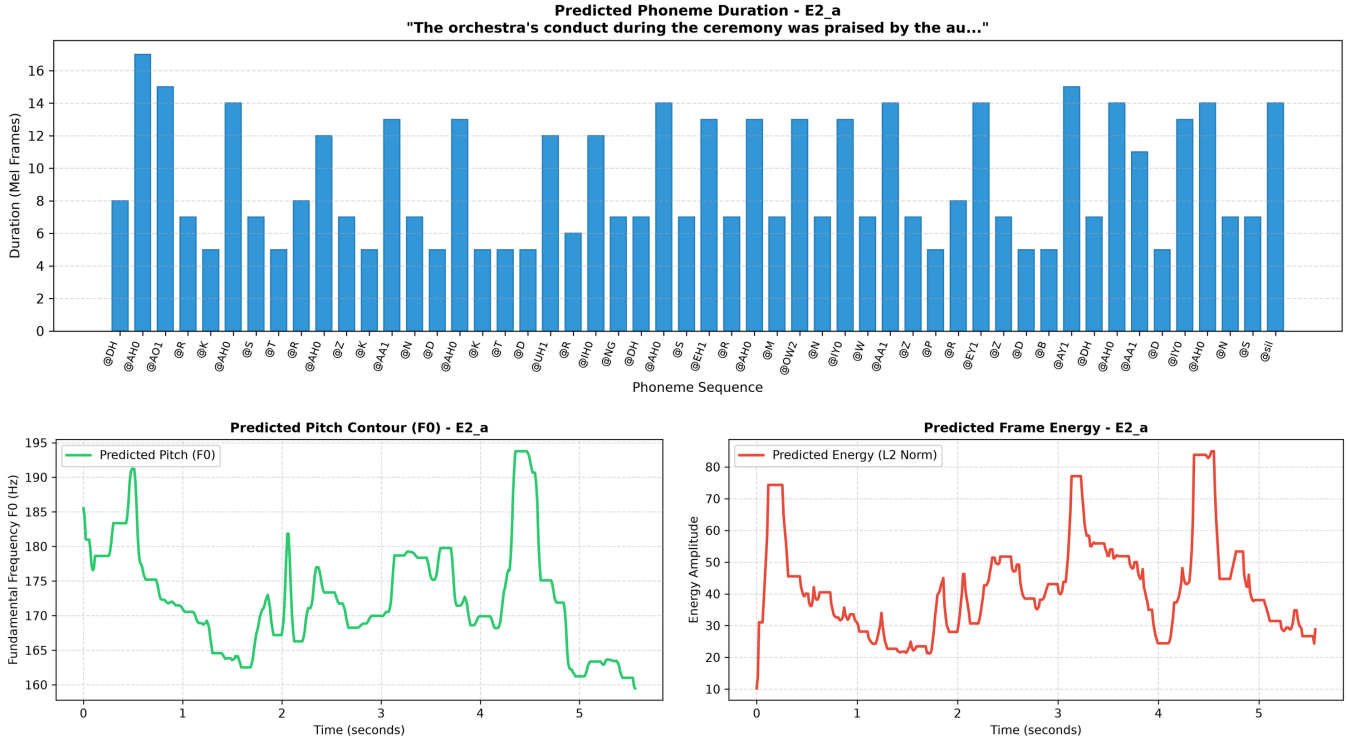


Figure 6. English E2a: predicted duration, pitch (F0) and energy.

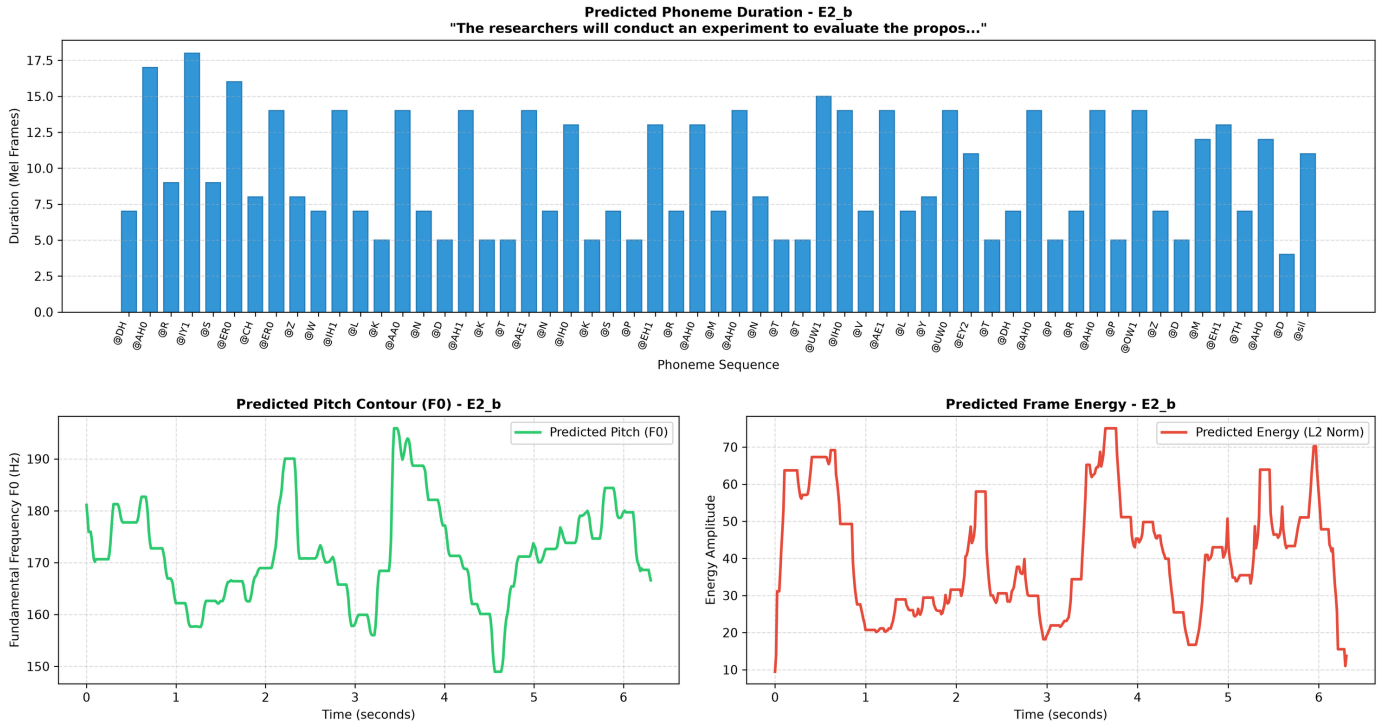


Figure 7. English E2b: predicted duration, pitch (F0) and energy.

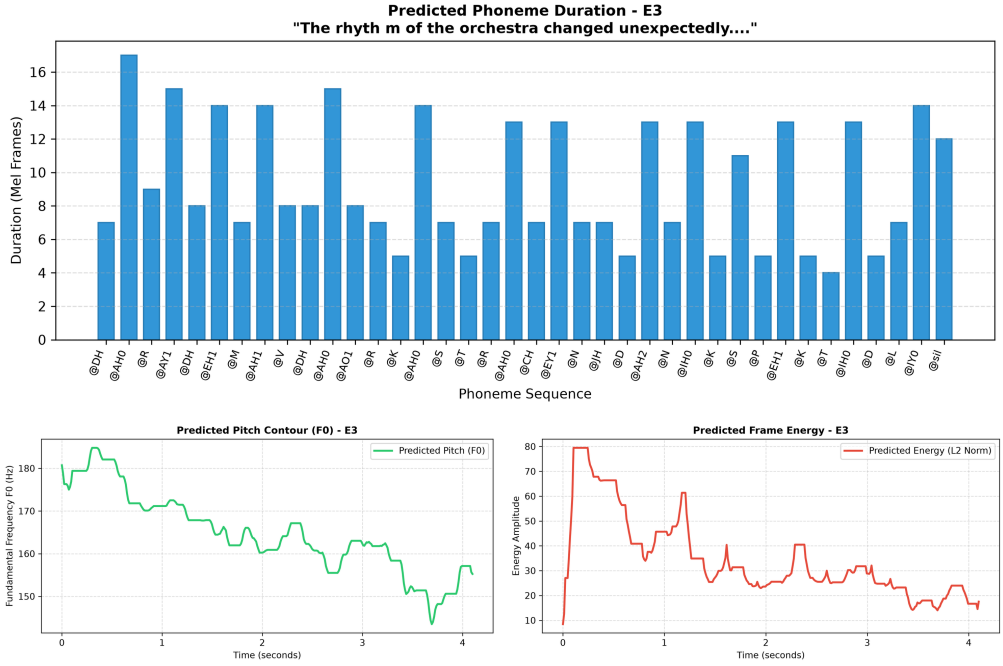


Figure 8. English E3: predicted duration, pitch (F0) and energy.

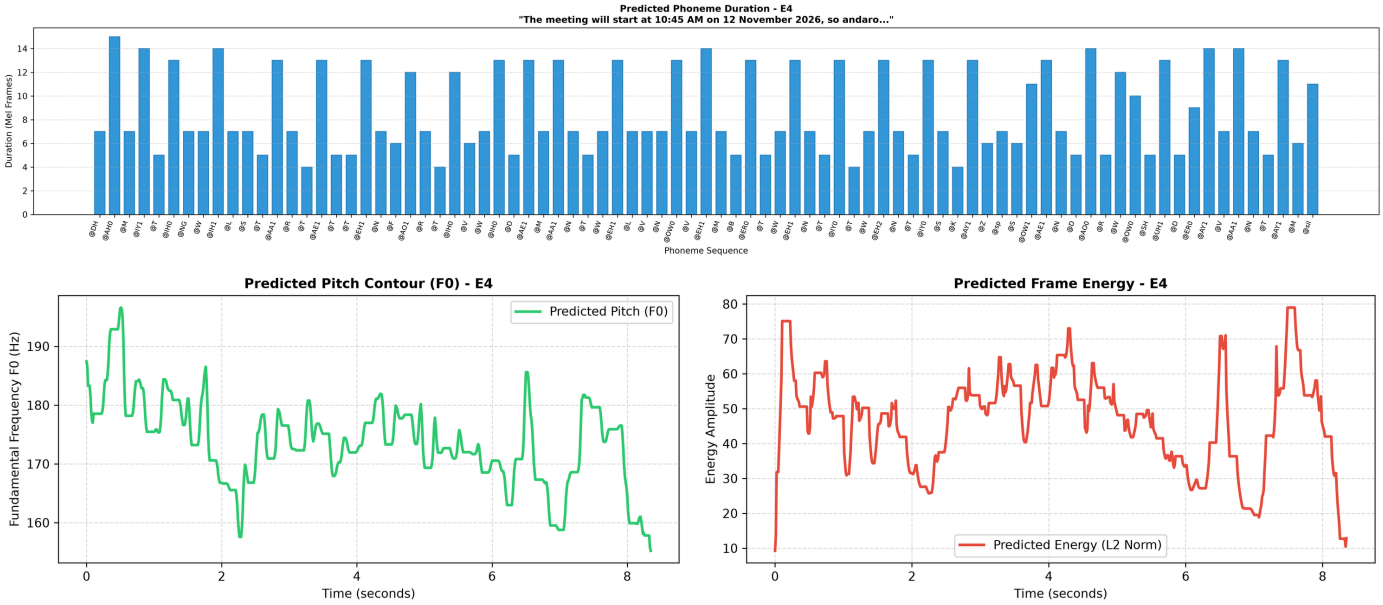


Figure 9. English E4: predicted duration, pitch (F0) and energy.

Reading the English plots. Pitch drifts downward across almost every sentence and finishes low, staying inside a narrow band of roughly 150 to 195 Hz around the training mean of 172.7 Hz. This flat trajectory occurs because the predictor learns the central tendency of the multi-speaker corpus rather than sentence-level prosodic accents, aligning with the rising validation pitch loss in Figure 2...

...The sharp energy rise in the first fraction of a second appears in every English utterance, representing an onset bias. E2a and E2b show identical structural trends despite contrasting syntactic roles for 'conduct'. In E4, digits and dates produce no localized pitch or energy inflection.

5.3 Telugu

T1 (4.95 s): pitch begins around 148 Hz, climbs rapidly to a peak of 288 Hz at 0.29 s, fluctuates between 160 and 220 Hz across the middle phrase, and drops to its lowest point of 120 Hz near the end (4.89 s). Energy shows multiple peaks with an initial high transient.

T2a (2.57 s): pitch starts near 187 Hz, reaches an early peak of 259 Hz at 0.21 s, hovers near 200–225 Hz, dips to 137 Hz around 2.47 s, and finishes near 191 Hz. Energy displays several distinct syllabic crests.

T2b (3.29 s): pitch begins at 164 Hz, peaks at 257 Hz at 0.55 s, remains sustained around 200–230 Hz, dips to 156 Hz near 3.11 s, and ends at 185 Hz. Energy begins high and gradually tapers off.

T3 (2.41 s): pitch starts around 183 Hz (not high), climbs swiftly to 287 Hz at 0.29 s, plunges steeply to 123 Hz near 1.03 s, recovers toward 230 Hz by 1.5 s, and terminates around 173 Hz. Energy initiates near 127 and slopes downward across the utterance.

T4 (8.72 s): pitch exhibits a jagged contour fluctuating between 122 Hz and 253 Hz (peak at 3.80 s), reaching its minimum of 122 Hz near 8.65 s. Energy drops close to zero around 6.8 to 7.2 s at the clause-separating comma, with a subsequent resurgence peaking around 8.2 s.

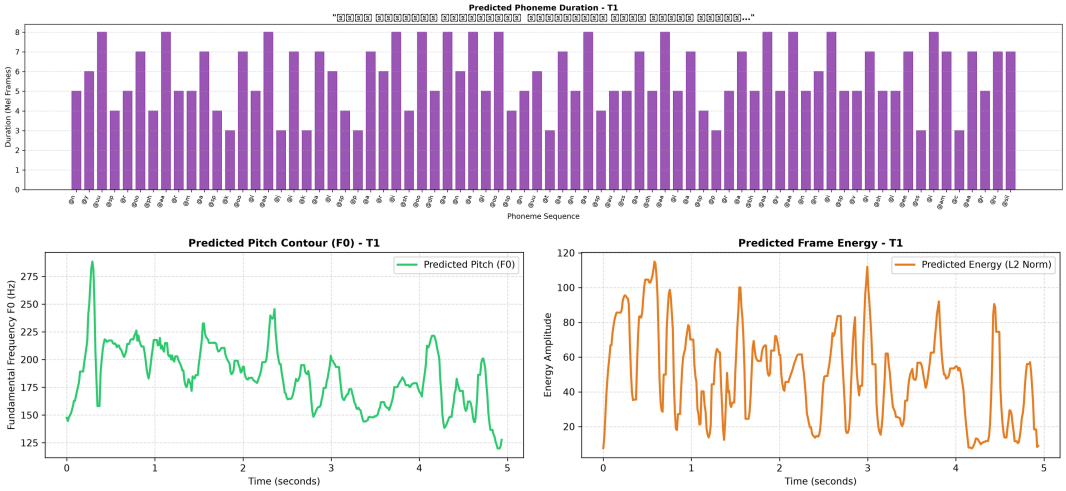


Figure 10. Telugu T1: predicted duration, pitch (F0) and energy.

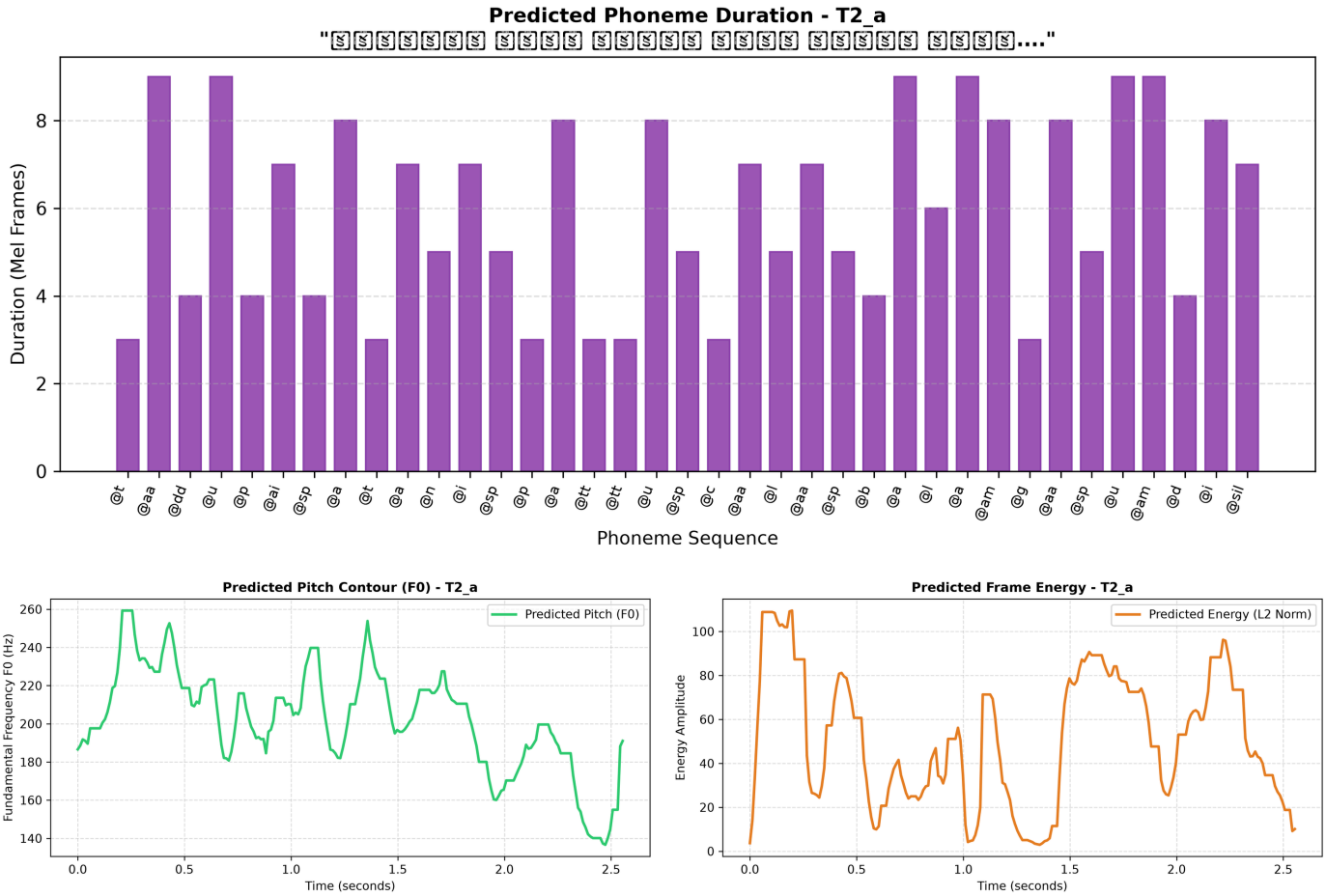


Figure 11. Telugu T2a: predicted duration, pitch (F0) and energy.

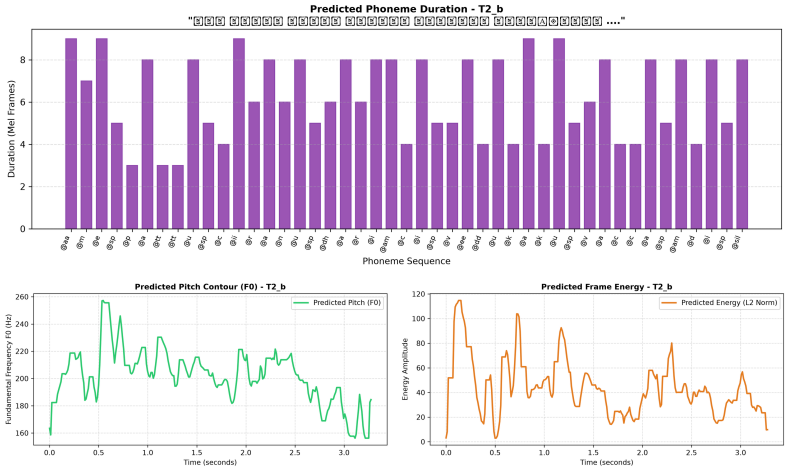


Figure 12. Telugu T2b: predicted duration, pitch (F0) and energy.

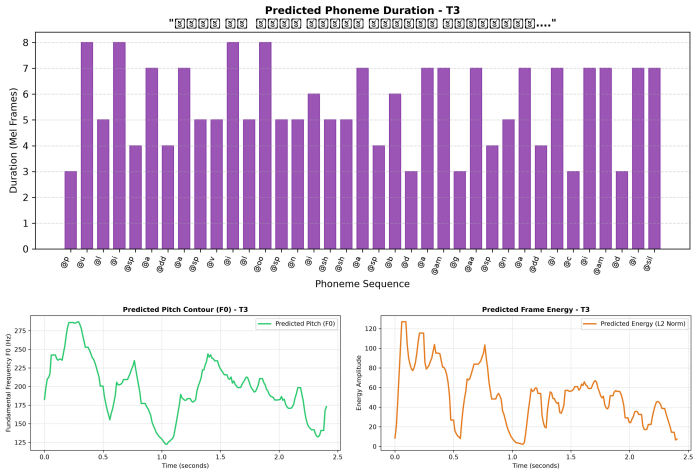


Figure 13. Telugu T3: predicted duration, pitch (F0) and energy.

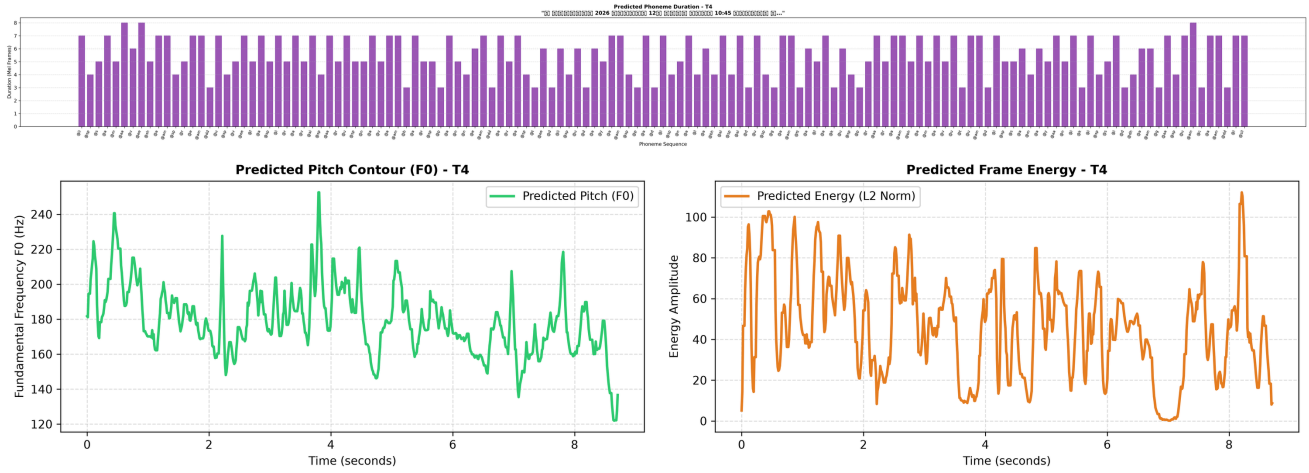


Figure 14. Telugu T4: predicted duration, pitch (F0) and energy.

Reading the Telugu plots. The Telugu pitch contours exhibit substantially wider dynamic range than English, reaching peaks near 290 Hz in T1 and T3 and troughing down to ~120 Hz. These predicted tracks fluctuate around the dataset training mean of 195.5 Hz (std 69.8 Hz, computed during preprocessing), reflecting a higher baseline and wider register than the English dataset (mean 172.7 Hz)...

...As in English, energy begins with an initial burst before settling into rhythmic fluctuations. In T4, energy plunges close to zero around 7.0 s, corresponding to the comma separator, while numbers and timestamps exhibit no anomalous discontinuities.

6. Comparing the two models, limitations and next steps

6.1 English and Telugu side by side

Telugu achieved lower mel and variance losses than English across all metrics. Two primary factors account for this: Telugu orthography provides a substantially more transparent grapheme-to-phoneme correspondence than English, and the IndicTTS corpus consists of clean single-speaker studio speech, whereas Svarah comprises multi-speaker crowd-sourced audio. However, both systems exhibit acoustic blurring from uniform duration targets. That points at shared methodological constraints—specifically duration heuristics and limited optimization steps—rather than language-specific obstacles.

Across the test sentences, both models handle structural complexities identically. The long compound sentences E1 and T1 generate extended sequences without discrete word-level transitions. The homograph pairs (E2a/E2b and T2a/T2b) reflect length differences purely proportional to token counts, without semantic pitch modulation. Date and time sentences (E4 and T4) run 8.4 to 8.7 s with flat prosody across unnormalized digits. Short sentences E3 and T3 are the cleanest outputs, though E3 lacks a break at the 'rhyth m' split, and T3 exhibits the most dynamic pitch excursion in the Telugu set (rising from 183 Hz to a 287 Hz peak at 0.29 s before plunging to 123 Hz at 1.03 s).

Metric	English	Telugu
Training mel loss, epoch 35	0.584	0.457
Best validation mel loss	0.852 (epoch 16)	0.684 (epoch 20)
Training duration loss, epoch 35	0.0187	0.0162
Validation pitch loss, epoch 35	0.615	0.479
Validation energy loss, epoch 35	2.959	2.254

6.2 Limitations

1. **Duration targets.** Frames were divided evenly across phonemes (T_mel // N_ph) rather than extracted from a forced aligner. Phoneme boundaries therefore do not align with acoustic transitions in the ground truth, encouraging the decoder to predict a blurry, time-averaged spectrogram. While this stands as the primary hypothesis for the acoustic degradation, it remains a hypothesis until verified through the diagnostic checks in Section 6.3.
2. **Training length.** Approximately 4,000 updates per model is orders of magnitude below standard FastSpeech 2 training budgets (typically exceeding 100k steps).
3. **Vocoder.** Pretrained HiFi-GAN (jaketae/hifigan-lj-v1) was trained on a single native English speaker (LJSpeech) and applied directly to Telugu and Indian-accented English without acoustic domain adaptation.
4. **Text front end.** Absence of numeral, calendar, and timestamp text normalizers leaves expressions in E4 and T4 unexpanded, while out-of-vocabulary words ('andaroo') rely on character-level fallback.
5. **Predictor overfitting.** Validation pitch and energy losses ascend steadily after epoch 10 while training losses decline, indicating overfitting to the training subset.
6. **Multi-speaker variance.** Splits were divided sequentially rather than speaker-stratified, and neither pitch nor energy was normalized on a per-speaker basis.
7. **Svarah dataset characteristics.** Svarah (Bhargav0044/svarah1) is an Indian English speech benchmark collected across multiple speakers, accents, and non-studio recording environments for ASR evaluation, rather than a curated single-speaker studio TTS dataset. Training a single-speaker FastSpeech 2 without conditioning on speaker embeddings forces the network to learn the central tendency across diverse vocal tracts and acoustic backgrounds. In contrast, IndicTTS Telugu is a single-speaker, studio-recorded dataset. This fundamental difference in acoustic consistency contributes significantly to the greater mel blurriness in English and explains a substantial part of the English vs. Telugu loss and quality disparity.

6.3 Checks I have not run yet

1. Resynthesize ground-truth mel-spectrograms with the HiFi-GAN vocoder. If resynthesized audio is degraded, the vocoder domain mismatch or STFT extraction parameters are primary culprits.
2. Plot predicted versus ground-truth mel-spectrograms for held-out validation utterances. Smeared harmonic striations and indistinct formants indicate under-training or alignment mismatch.
3. Validate spectrogram dynamic ranges and normalization bounds to rule out scale disparities.

6.4 What I would try next

- Integrate Montreal Forced Aligner (MFA) to generate accurate phonetic alignments.
- Extend training to 100k+ iterations using warm restarts and decayed learning rates.
- Implement speaker embeddings or restrict training to high-SNR single-speaker subsets.
- Implement comprehensive text normalization modules for numerals, dates, and times.
- Fine-tune language-adapted vocoders matched to the target acoustic domains.

7. Reproducibility

7.1 Software

Python 3, PyTorch (CUDA build for training), librosa, soundfile, scipy, numpy, PyYAML, matplotlib, tqdm, and speechbrain (for HiFi-GAN). Complete pinned dependencies are documented in `requirements.txt`.

7.2 Commands

Preprocessing:

```
python English/preprocess.py
python Telugu/preprocess.py
```

Training (35 epochs each):

```
python English/train.py --config English/config.yaml --epochs 35
python Telugu/train.py --config Telugu/config.yaml --epochs 35
```

Inference:

```
python English/inference.py --config English/config.yaml --checkpoint English/checkpoints/best_model.pth
python Telugu/inference.py --config Telugu/config.yaml --checkpoint Telugu/checkpoints/best_model.pth
```

Configuration files are located at `English/config.yaml` and `Telugu/config.yaml`. Complete epoch-by-epoch loss trajectories are archived in `Results/<Language>/training_history.json`.

Code Repository: github.com/LightningHat/FastSpeech-2-Based-Text-to-Speech-Synthesis

Trained Model Checkpoints: Google Drive Checkpoint Folder